Public Economics
Practicum 6

Discussion Questions Rosen:

	all editions
Chapter 16	1, 7, 8

Additional Exercises:

6.1 The supply of both good X and good Y is perfectly elastic at $P = 10$. Suppose the demand for good X can be represented by:

$$Q_X = 22 - \frac{1}{4}P_X$$

and the demand for good Y by:

$$Q_Y = 50 - P_Y$$

(a) Find the elasticity of demand for both good X and good Y in equilibrium.

(b) Suppose that an ad valorem tax is placed on both goods. The government needs to collect a given amount of tax revenue on the markets for good X and good Y, and aims to minimise the total loss in efficiency due to taxation. Good Y is taxed at a rate of 5%. What must be the rate at which good X is taxed?

(c) Now suppose the poor people consume relatively many units of good X while rich people consume relatively many units of good Y. Also, suppose that the government maximises social welfare, which is given by the sum of the utilities of all citizens. Poor people have a higher marginal utility of income than rich people. How would this effect the optimal tax rates?

6.2 Consider two goods (say, soap and shampoo) with identical demand functions: $P = 10 - Q$. Suppose supply of both soap and shampoo is perfectly elastic at the same price level: $P = 2$. Suppose one of the two goods is currently taxed at $t = 2$ per unit.

(a) What is the revenue of this tax? What is the excess burden?

Now suppose both goods are taxed at $t = 1$ per unit.

(b) What is the total revenue of these taxes? What is the total excess burden?

(c) From an economic efficiency point of view, which of these two tax schemes is preferable?

6.3 The Erasmus University considers placing baskets with flowers on the Woudestein campus. There is place for at most 100 baskets. After careful investigation, the EUR has learned that students' (total) benefit of these flower baskets is equal to $2F$, where F is the number of flower baskets. The cost of each basket is 1. The EUR wants to pay for the flowers by charging a unit tax t_c on each cup of coffee sold. Demand for coffee can be represented by $Q_c = 300 - 100P_c$. The supply of coffee is perfectly elastic at $P_c = 1$.

(a) If we ignore the excess burden of the tax, how many flower baskets are placed?

(b) What level of unit tax on coffee is necessary to cover the cost of planting the flowers? What is the excess burden of this tax? How large is the increase in campus welfare?

(c)* Show that for any number of F, the *marginal* cost of a flower basket, taking account of the excess burden of the coffee tax, equals $MC = 1 + \frac{t_c}{2 - 2t_c}$. (Hint: use the chain rule: $\frac{\partial EB_C}{\partial F} = \frac{\partial EB_c}{\partial t_c} \cdot \frac{\partial t_c}{\partial F}$. Also, remember that $\frac{\partial t_c}{\partial F} = \frac{1}{\partial F / \partial t_c}$)

(d) What is the optimal number of flower baskets and the optimal level of the unit tax? What is the excess burden of this tax? How large is the increase in campus welfare?

6.4 (this is an old exam exercise with propositions, apart from e and f)

A small city has 50 identical poor citizens and 50 identical rich citizens. On a plot of land in the city, a park can be constructed. The cost of the park are 10.000 euro. Each rich inhabitant values the park at 150 euro, and each poor inhabitant values the park at 50 euro. The cost of constructing the park will be covered through two new taxes. First, a part of the cost of constructing the park will be financed through a unit tax on opera tickets equal to 5 euro per ticket, levied on the producers. Second, the remaining cost will be covered through a head tax, such that each citizen pays the same amount in this head tax.
Only the rich citizens go to the opera. The demand for opera tickets by all rich inhabitants together is given by:

$$P = 100 - \frac{1}{20}Q$$

Marginal cost of opera tickets is constant. In the absence of a tax on opera tickets, supply of opera tickets is given by $P = MC = 20$.
The social welfare function of the city is additive:

$$SW = 50U_R + 50U_P$$

where U_R is the utility of a rich person and U_P is the utility of a poor person. The marginal utility of income of a rich person equals 1. The marginal utility of income of a poor person equals α.

(a) The Excess Burden of the tax on opera tickets equals 250 euro.

(b) If the demand for opera tickets would be more elastic, tax revenue would be smaller.

(c) Each rich individual would pay in total 175 euro in taxes if the park would be constructed.

(d) If $\alpha = 1.1$, building the park increases social welfare.

(e) Explain how the outcome at d) relates to the trade-off between efficiency and distribution.

(f) Suppose that only the poor would go to the opera (or that another good would be taxed, only consumed by poor, with demand as stated above). For which level of α would it be optimal to build the park?

6.5 Consider a village with 50 identical rich inhabitants and 50 identical poor inhabitants. The village considers investing in a public good, which costs 3000 euro. Each inhabitant of the village values the public good at V euro. The public good is financed by a tax on good X. Demand for good X can be given by $P = 100 - \frac{2}{5}Q$, where P is the price per unit in euro and Q is quantity. Supply can be given by $P = 10 + \frac{1}{5}Q$. The government implements a unit tax of 30 euro, to be paid by producers of good X.
Only poor inhabitants consume good X. Only rich inhabitants produce good X. The social welfare function of the village is additive:

$$SW = 50U_R + 50U_P$$

where U_R is the utility of a rich person and U_P is the utility of a poor person. The marginal utility of income of a rich person equals 1. The marginal utility of income of a poor person equals $\alpha > 1$.

1. **(a)** Derive the share of the tax burden that is borne by the consumers of good X.

(b) Explain why the rich would prefer a tax on good X over a head tax.

1. **(c)** Suppose $V = 40$. Derive for which levels of α it is socially optimal to invest in the public good given that the investment is financed by the tax on good X. Explain

   (d) How would your answer to (c) change if the poor had a larger weight in the social welfare function than the rich?

Answers to exercises week 6

Rosen 16.1

The commodities should optimally be taxed according to the inverse elasticity rule. Using the numbers in exercise, we get

$$\frac{t_{cable}}{t_{satellite}} = \frac{\eta_{satellite}}{\eta_{cable}} = \frac{7.4}{0.51} = 14.5$$

Thus, tax rates on basic cable should be 14.5 times higher than tax rates on satellite television because basic cable is inelastically demanded, while demand for satellite television is highly elastic.

Among the assumptions that go into the inverse elasticity rule are that goods are neither complements nor substitutes. In this case, it is likely that the first of these assumptions is false – basic cable and satellite television are likely substitutes for each other.

Rosen 16.7, 8th and 9th edition

The problem with a one-time tax on profits is that the government has an incentive to renege on its promise. The time inconsistency of optimal policy occurs when the government cannot implement an optimal tax policy because the stated policy is inconsistent with the government's incentives over time. Since pharmaceutical companies know that government has an incentive to impose a tax on profits more than once, their behavior will be affected by the tax, creating an excess burden.

Rosen 16.7, 10th edition

The problem is that the R&D cost are incurred before the drugs and vaccins can be sold. After development, international organizations may put pressure on the pharmaceuticals to sell the drugs at relatively low cost (a bit above marginal cost), which would make it impossible to cover the (sunk) R&D cost. The optimal policy would be a promise to pay a price sufficiently high to cover the R&D cost, but this promise is not credible: there is a time-inconsistency problem in the sense that after the development, the international organizations are better off by reneging on their promise and demand lower prices.

There are no easy solutions: Fixing prices ex ante (before development) is difficult, as true cost are hard to forecast before development. A fixed mark-up (bove marginal cost) is also problematic, is it removes incentives to make efficient cost-reducing investments by the pharmaceuticals. In practice, international organizations sponsor R&D. Another solution is patents, which the pharmaceuticals can use to reduce competition by copy-cats.

Rosen 16.8

a) True. To illustrate, consider the simplest example where there are only two

goods: consumption goods and leisure. The budget constraint is equal to: $p_C C + wL = I$, where p_C and w are the prices of consumption goods and leisure, C and L are the quantities of consumption and leisure, and I is income. Then a proportional tax τ on all goods changes the budget constraint to: $(1+\tau)p_C C + (1+\tau)wL = I$, or rearranging, $p_C C + wL = I'$, where $I' = \frac{I}{(1+\tau)} < I$. Thus, a proportional tax on all goods does not change relative prices and is equivalent to taking away time (or earnings capacity). So it is equivalent to a lump sum tax.

b) False. If leisure is not taxed, generally efficiency requires different tax rate for different goods (Ramsey's inverse elasticity rule). Imagine a more complicated budget constraint: $p_C C + p_F F + wL = I$. If leisure cannot be taxed, then a tax on commodities τ leads to a budget constraint of $(1+\tau)p_C C + (1+\tau)p_F F + wL = I$, which does change the relative price of leisure compared with food or consumption goods. Thus, it is not a lump sum tax. Instead, the inverse elasticity rule given in equation (16.9) would suggest that the ratio of the tax rates are inversely related to the ratio of the compensated demand elasticities for all commodities that can be taxed. That is, $\tau_C/\tau_F = \eta_F/\eta_C$.

c) True. See Figure 16.3 in the book.

d) One notion of horizontal equity is that people in equal positions should be treated equally by the tax system. Under this traditional notion of horizontal equity, the fact that Tom's workplace provides free access to a fitness room suggests this kind of compensation should be taxed; Jerry pays "full taxes" on his compensation while Tom does not. Another notion of horizontal equity relies on the *utility definition* of horizontal equity. This concept says that if two individuals have the same utility without taxes, they should have the same utility with taxes, and the taxes should not affect the utility ordering. One implication of the *utility definition* is that any existing tax structure does not violate the notion of horizontal equity if individuals are free to choose their activities and expenditures. If Tom and Jerry have free choice between the two different jobs (and identical preferences), then the net after-tax rewards (including amenities) must be the same at both jobs; otherwise there would be migration. In this case, the before-tax wage on Tom's job adjusts for the fact that there is a fringe benefit.

Exercise 6.1

a) $\varepsilon_x = \left|\frac{\partial Q_X}{\partial P_X}\frac{P_X}{Q_X}\right| = \left|-\frac{1}{4}\frac{10}{19\frac{1}{2}}\right| = 0.128$. Similarly, $\varepsilon_Y = \left|-1\frac{10}{40}\right| = 0.25$.

b) This implies the govenrment should use the inverse elasticity rule: $\frac{t_X}{t_Y} = \frac{\varepsilon_Y}{\varepsilon_X}$. Substituting gives $\frac{t_X}{0.05} = \frac{0.25}{0.128}$. Hence, $t_X = 0.0975$ (9.75%).

c) This implies that a euro tax revenue collected on the market for good X reduces social welfare more than a euro tax revenue collected on the markt for good Y. The inverse elasticity rule only considers efficiency, and does not incorporate any

redistributional preferences. Hence, if the government wants to redistribute from rich to poor, it should adjust the tax rates (decrease in t_x, increase in t_y), yielding $\frac{t_X}{t_Y} < \frac{\varepsilon_Y}{\varepsilon_X}$.

Exercise 6.2

a) Without tax, $Q = 8$. With $t = 2$, $Q = 6$. Tax revenue $2 \cdot 6 = 12$. Excess burden $= \frac{1}{2}t \cdot \Delta Q = \frac{1}{2}2 \cdot 2 = 2$.

b) With $t = 1$, $Q = 7$. Hence, tax revenue on each good is 7, which makes total tax revenue 14. Excess burden on each good is $\frac{1}{2}1 \cdot 1 = \frac{1}{2}$. Hence, total excess burden is 1.

c) The second scheme yields both more revenue and less excess burden, so is preferable (from an efficiency point of view) to the first scheme.

Exercise 6.3

a) Marginal benefit of a basket $MB_F = 2$. Marginal cost without excess burden is 1. So all 100 places will be filled with flower baskets.

b) Tax revenue is given by $t_c Q_c = t_c \left[300 - 100(P_c + t_c)\right] = 200t_c - 100t_c^2$. (note that $P_c = 1$). Total cost of 100 baskets is 100, so it must be that $200t_c - 100t_c^2 = 100$, or $t_c^2 - 2t_c + 1 = 0$. This gives $t_c = 1$.

Excess burden is given by $\frac{1}{2}t \cdot \Delta Q_c$. Without tax, $Q_c = 200$. With $t_c = 1$, $Q_c = 100$. So, EB $= \frac{1}{2}1 \left[200 - 100\right] = 50$. Benefit of flowers $2F = 200$. Direct cost is 100. Excess burden is 50. Hence, welfare has increased by $200 - 100 - 50 = 50$.

c) The cost of placing F baskets is F. Hence, to place F baskets, t_c must be such that tax revenue $t_c Q_c = F$. Derived above, $t_c Q_c = 200t_c - 100t_c^2$. This gives $F = 200t_c - 100t_c^2$. Using $\frac{\partial t_c}{\partial F} = \frac{1}{\partial F / \partial t_c}$, we can derive:

$$\frac{\partial t_c}{\partial F} = \frac{1}{200 - 200t_c}$$

This gives how t_c changes with F.

The excess burden of t_c equals $\frac{1}{2}t_c \Delta Q_c$. Without tax, $Q_c = 200$. With tax t_c, $Q_c = 200 - 100t_c$. Hence, $\Delta Q_c = 200 - \left[200 - 100t_c\right] = 100t_c$. Substituting this gives EB $= 50t_c^2$. So F has effect on EB_c through t_c (a change in F affects t_c, which in turn affects EB_c). In math, the marginal effect of F on EB_c is given by: $\frac{\partial EB_C}{\partial F} = \frac{\partial EB_c}{\partial t_c} \cdot \frac{\partial t_c}{\partial F}$. This gives:

$$\frac{\partial EB_C}{\partial F} = 100t_c \cdot \frac{1}{200 - 200t_c} = \frac{t_c}{2 - 2t_c}$$

So, the direct cost of an additional basket is 1, and the associated increase in excess burden is $\frac{t_c}{2-2t_c}$. Hence, total marginal cost (the marginal cost of public funds) is $1 + \frac{t_c}{2-2t_c}$. (Note: in the context of government cost-benefit analysis, this expression gives the *marginal cost of public funds*).

d) Marginal benefit of F is 2. Marginal direct cost is 1. Marginal excess burden is derived above. Hence, the optimal level of F is given by:

$$2 = 1 + \frac{t_c}{2 - 2t_c}$$

This can be solved for $t_c = \frac{2}{3}$.
Substituting $t_c = \frac{2}{3}$ into $F = 200t_c - 100t_c^2$ gives $F = \frac{800}{9}$. Excess burden is $\frac{1}{2}\frac{2}{3}\left[200 - 133\frac{1}{3}\right] = 22\frac{2}{9}$.
Welfare improvement is now $2 \cdot \frac{800}{9} - \frac{800}{9} - 22\frac{2}{9} = 66\frac{2}{3}$.

6.4
a) True. Before the tax, the equilibrium is given by $100 - \frac{1}{20}Q = 20 \rightarrow Q = 160\dot{0}$. With tax, this becomes $100 - \frac{1}{20}Q = 20 + 5 \rightarrow Q = 150\dot{0}$. Hence, the Excess Burden equals $\frac{1}{2}5\,(1600 - 1500) = 250$. Beyond the question: this is 5 euro per rich inhabitant.
b) True. Equilibrium quantity would go down more, as demand is more price-sensitive. This reduces tax revenue.
c) True. Total opera tax revenue equals 5*1500 = 7500, which is 150 per rich inhabitant. The remaining 2500 euro is collected through a lumpsum tax, which is 2500/100 = 25 euro per person. Hence, rich inhabitants pay 175 euro in taxes for the park.
d) False. The park affects each rich inhabitant's net utility by $150 - 175 - 5 = -30$ euro. Note that the -5 comes from the Excess burden of the opera tax. It affects each poor inhabitant's net income by $50 - 25 = 25$ euro, which corresponds to $25 * \alpha$ extra utility. Hence, constructing the park leads to higher SW when

$$50 * -30 + 50\alpha 25 \geq 0 \rightarrow \alpha \geq 1.2$$

e) The economic cost of the park are 10250 euro. The park is valued at 7500 + 2500 euro (ignoring the tax, each rich citizen's willingness to pay for the park equals 150 and each poor citizen's willingness to pay is 50). Hence, building the park is not efficient. However, given the tax on opera, the park also redistributes from rich to poor. If distributional motives are sufficiently strong (here: the difference in marginal utility between rich and poor sufficiently high), social welfare can increase despite the reduction in efficiency.
Beyond the question: a Pareto-improvement is theoretically possible: if, rather than building the park, each rich citizen would give an amount between 25 and 30 euro to a (different) poor person, everyone would be better off.
f) Now, the park affects each rich inhabitant's net utility by $150 - 25 = 125$ euro. It affects each poor inhabitant's net utility by $50 - 175 - 5 = -130$ euro. Hence,

constructing the park leads to higher SW when

$$50 * 125 - 50\alpha 130 \geq 0 \rightarrow \alpha < \frac{6250}{6500}$$

As marginal utility of poor is generally larger than that of rich, α will be larger than one. Hence, it is not optimal to build the park.
Note that this exercise shows that: (i) you need to take the excess burden into account when determining the efficiency / social welfare effects of investments, and (ii) that for social welfare, it matters who is effectively bearing the cost of taxation.

Exercise 6.5
a) Equilibrium without tax: $100 - \frac{2}{5}Q = 10 + \frac{1}{5}Q \rightarrow Q = 150$ and $P = 40$. Equilibrium with tax $100 - \frac{2}{5}Q = 10 + 30 + \frac{1}{5}Q \rightarrow Q = 100$, consumers pay 60 and producers keep 30. Hence, consumers bear 2/3 of the tax burden. Note: tax revenue is 3.000 euro.
b) In case of the tax on X, there is an excess burden. The excess burden is $\frac{1}{2}30\,(150 - 100) =$ 750 euro. Hence, total cost are 3750 euro. The rich are able to shift 2/3 of these total cost to the poor. Hence, the cost to each rich person equals 1250 / 50 = 25 euro. In case of a head tax, there is no excess burden and rich and poor would share the tax burden equally: all inhabitants would pay 30 euro. Hence, the rich prefer the tax on X, despite the excess burden.

c) The rich bear 1/3 of the total cost of the tax, the poor bear 2/3. Hence, it is socially optimal to invest if $50 * 40 + 50 * 40\alpha \geq 1250 + 2500\alpha \rightarrow 750 \geq 500\alpha \rightarrow \alpha \leq 1.5$. The investment benefits the rich, but hurts the poor. Total benefits are larger than total cost. However, if society values a euro to the poor sufficiently more than a euro to the rich (here: because the marginal utility of income of a poor person is sufficiently higher), it is socially optimal to refrain from making the investment.
d) Intuitively: if the poor have a larger weight in the SW-function, society values a euro to the poor even more than before. This makes it socially less attractive to invest, such that the investment is done only for sufficiently low values of α. In math: Suppose the SW-weight of the poor is $w > 1$. it is socially optimal to invest if $50 * 40 + w * 50 * 40\alpha \geq 1250 + w * 2500\alpha \rightarrow 750 \geq 500\alpha w \rightarrow \alpha \leq \frac{1.5}{w}$, which decreases in w.